Abdullahi Ahmad

Machine Learning Engineer · AI Research — Large Language Models, Agentic AI, Computer Vision

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RESEARCH PROFILE

Machine Learning Engineer and early-career AI researcher with over four years of experience designing, evaluating, and deploying deep learning, large language model (LLM), and computer vision systems. Co-author of a peer-reviewed performance–efficiency benchmark of transformer models for low-resource NLP (accepted, AIATSI 2025) and Grand Prize winner of the internationally contested 2023 Huawei ICT Competition Global Final (Innovation/AI track). Hands-on research experience spans agentic multi-agent LLM frameworks (LangChain, LangGraph), Retrieval-Augmented Generation, behavioural recognition from video, and temporal modelling of network telemetry.

RESEARCH INTERESTS

AI for Cybersecurity and Behavioural Threat Analytics; Agentic AI and Multi-Agent LLM Systems; Large Language Models and Retrieval-Augmented Generation; Graph Representation Learning and Knowledge Graphs; Machine Learning and Deep Learning; Network and IoT Security; Natural Language Processing (including Low-Resource Languages); Computer Vision and Behaviour Recognition.

EDUCATION

B.Eng., Computer Engineering — Upper Second Class (CGPA 3.69 / 5.00) Oct 2018 – Nov 2024

Ahmadu Bello University, Zaria, Nigeria

- Supervisor: Prof. Muhammad Bashir Muazu.
- Relevant coursework: Machine Learning, Optimization, Digital Signal Processing, Control Systems, Embedded Systems.

Diploma, Computer Engineering (CGPA 3.52 / 4.00) Aug 2016 – Nov 2018

Institute of Computing and ICT, Ahmadu Bello University, Zaria, Nigeria

- Supervisor: Mal. Nura Dari.

HONOURS & AWARDS

- **Grand Prize Winner — Huawei ICT Competition, Global Final, Innovation/AI Track (May 2023).** First place in the AI/Innovation track of Huawei's flagship international ICT competition, contested by university teams worldwide.
- **Grand Prize — Huawei ICT Competition, Regional Final (Southern Africa), Innovation/AI Track (Mar 2023).**

PUBLICATIONS

Peer-Reviewed Conference Papers — Accepted (Forthcoming)

Muazu, M. B., Bayero, A., **Ahmad, A.**, Ibrahim, S., & Dahir, D. I. (2025). A performance-efficiency benchmark of transformer models for code-mixed Hausa sentiment analysis. *AIATSI 2025: Artificial Intelligence – Advanced Topics and Societal Innovation*, Dakhla, Morocco, November 5–7, 2025. (Accepted.)

Posters & Departmental Presentations

Abdullahi, A. (2024). An artificial intelligence smart waste robot [Poster presentation]. Department of Computer Engineering, Ahmadu Bello University, Nigeria.

Manuscripts in Preparation (not yet peer-reviewed)

Abdullahi, A. A multi-agent LLM framework for automating the software development lifecycle.

Abdullahi, A. An LLM-powered multi-agent coding agent with LangGraph for code generation and analysis.

Abdullahi, A. A multi-agent LLM system for startup analysis using LangGraph.

Abdullahi, A. NeuroScan: Brain tumor classification using deep learning.

Abdullahi, A. An LLM-based adaptive learning system for personalized educational pathways.

Abdullahi, A. Heart disease classification using Random Forest and Support Vector Machine algorithms.

SELECTED RESEARCH PROJECTS

Multi-Agent LLM Frameworks with LangChain and LangGraph Ongoing

- Designed and implemented agentic LLM frameworks in which role-specialized agents coordinate over shared state to automate multi-step workflows, including the software-development lifecycle, code generation and analysis, and startup analysis.
- Investigated agent-orchestration patterns, tool integration, and multi-step reasoning with LangGraph; this work forms the basis of three first-author manuscripts in preparation.

Network Failure Prediction on 5G Core CNFs using eBPF Jul – Aug 2022

Technologies: Python, LSTM, eBPF, 5G Core (Containerized Network Functions)

- Investigated early prediction of failures in Containerized Network Functions (CNFs) on a 5G core, where faults must be anticipated from low-level network behaviour before service degradation occurs.
- Co-developed an LSTM-based temporal model over low-overhead eBPF telemetry and optimized the architecture to reach 88.3% prediction accuracy, improving early detection of network faults.
- Research significance: demonstrates extraction of predictive signal from kernel-level network telemetry — a methodological foundation for behavioural anomaly detection in network security.

CompengChat — RAG Assistant for Institutional Question Answering Jul 2023

Technologies: RAG, LangChain, LLMs, Telegram Bot API, Web Deployment

- Designed and deployed a Retrieval-Augmented Generation framework combining dense vector retrieval with LLM inference to answer student and public queries over an institutional knowledge base.
- Integrated the system into the departmental website and a Telegram bot, improving access to course, lecturer, and departmental information and reducing repetitive administrative inquiries.
- Research significance: demonstrates vectorized knowledge integration for grounding LLM outputs in retrieved evidence.

SmartWaste — Autonomous Waste-Sorting Robot Mar – Jul 2023

Technologies: Deep Learning (CNN), Computer Vision, Robotic Arm, Raspberry Pi

- Led development of an autonomous perception–actuation system in which a CNN-based vision model detects non-biodegradable waste and a robotic arm sorts it into the correct bin.
- Implemented and validated the end-to-end prototype on Raspberry Pi; results formed the basis of an accepted poster presentation at Ahmadu Bello University.

Brain Tumour Classification via Transfer Learning Jan 2024

Technologies: Python, TensorFlow / Keras, VGG16, Transfer Learning

- Developed and evaluated a transfer-learning pipeline using VGG16 as a feature extractor to classify brain tumours from MRI images, achieving 92.3% accuracy.
- Research significance: explores label-efficient deep learning for high-stakes medical classification; related manuscript (NeuroScan) in preparation.

Medical Knowledge Graph RAG System Nov – Dec 2025

Technologies: Qdrant, Neo4j, OpenAI Embeddings, GPT-4, RAGAS, Gradio, Docker

- Built a hybrid RAG system combining vector retrieval and graph traversal over 39,000+ UMLS concepts enriched with PubMed literature and evidence-based citations.
- Integrated GPT-4 tool calling for graph enrichment and concept retrieval; implemented RAGAS evaluation and deployed a Gradio interface with Docker-based Qdrant services.

AI-Powered Phishing Email Detector Sep 2025

Technologies: Python, Pandas, Scikit-Learn, XGBoost, Streamlit

- Developed an email-classification pipeline using text and metadata features; compared Naive Bayes, Linear SVC, and XGBoost models.
- Deployed the trained XGBoost model through a Streamlit application for interactive phishing detection.

End-to-End Customer Segmentation Strategy Platform Aug – Sep 2025

Technologies: Python, K-Means, Random Forest, Streamlit

- Built a customer analytics platform using K-Means to identify five behavioural customer segments and Random Forest to classify new customers.
- Developed a multi-page Streamlit dashboard supporting profitability modelling, segment analysis, and customer predictions.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Mar 2026 – Present

TerraHaptix (Autonomous Robotics & Drones) · Full-time, On-site · Abuja, Nigeria

- Designed, developed, and optimized production-grade machine learning models for autonomous robotics, improving model accuracy while validating reliability under real-time operational constraints.
- Investigated emerging AI techniques in computer vision and model optimization; prototyped and benchmarked candidate solutions that enhanced platform capabilities and informed product-roadmap decisions.
- Collaborated with data-engineering and software teams to design and maintain scalable data pipelines for machine learning workloads.

Machine Learning Engineer Intern Jan 2023 – May 2024

Nafabat.AI · Remote · California, USA

- Fine-tuned LlamaVID, a video–language model, on a large-scale CCTV dataset to recognize shoplifting behaviour, enabling automated behavioural threat detection across multiple camera feeds.
- Designed and evaluated a Retrieval-Augmented Generation (RAG) system for natural-language querying of time-series telemetry in InfluxDB, reducing manual data-lookup time.
- Implemented and deployed an end-to-end RAG application with a Gradio interface, delivering interactive, LLM-powered document and data querying to end users.

Network Engineer Intern Nov 2024 – Apr 2025

Cosmopolitan University, Abuja · Abuja, Nigeria

- Deployed and configured campus physical-security infrastructure, including Dahua CCTV surveillance at key locations (library, main gate, auditorium) and ZKTeco biometric attendance systems for staff and students.
- Configured enterprise wireless and hostel network infrastructure (Huawei AirEngine 5761R-11 access points) and supported the Head of IT with technical maintenance and equipment inventory for the Computer Engineering Lab and Maker Space.

INVITED TALKS & PRESENTATIONS

- “AI Assistive Glasses for the Visually Impaired” — Project presentation, Nigeria Computer Society AI Summit.

TEACHING, MENTORSHIP & OUTREACH

Tutor — Machine Learning Fundamentals & Optimization (COEN417, COEN505) Feb 2024

Ahmadu Bello University, Zaria, Nigeria

- Delivered tutorials on machine learning fundamentals and optimization to undergraduate engineering students, translating formal concepts into accessible instruction.

ICT Instructor — ICT Bootcamp Feb 2024

National Association of Jigawa State Students, Nigeria

Instructor — Python Bootcamp for Kids Dec 2019

Department of Computer Engineering, Ahmadu Bello University, Zaria, Nigeria

- Organized AI and robotics workshops for youth empowerment in Babura LGA and mentored aspiring engineers on building projects with AI and robotics.
- Shared applied knowledge of LangChain and generative AI with local and global developer communities.

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, Rust.

Backend & API Frameworks: FastAPI, Flask.

Frontend & UI Frameworks: React.js, Streamlit, Gradio, Tailwind CSS.

Machine Learning: Scikit-Learn, XGBoost, Apache Spark MLlib; experimental design, evaluation, and benchmarking.

Deep Learning: PyTorch, TensorFlow, Keras, MindSpore; CNNs, LSTMs, transformer architectures, transfer learning.

LLMs & Agentic AI: LangChain, LangGraph, Retrieval-Augmented Generation, multi-agent systems, LLM fine-tuning, prompt engineering, vector retrieval, Gradio.

Natural Language Processing: Transformer benchmarking, sentiment analysis, low-resource and code-mixed text processing.

Computer Vision: Image classification, object and behaviour detection, video understanding (LlamaVID), VGG16 transfer learning.

Networking & Security Systems: eBPF telemetry, 5G core CNFs, CCTV surveillance (Dahua), biometric access systems (ZKTeco), enterprise Wi-Fi (Huawei AirEngine).

Data Engineering & Databases: MySQL, InfluxDB, HBase, Hive, Hadoop, Spark, Huawei MapReduce Service.

Cloud Platforms: AWS, Google Cloud Platform, Microsoft Azure, Huawei Cloud.

Robotics & IoT: ROS, Raspberry Pi (Zero, 4), Arduino, ESP32, robotic-arm integration.

Research & Development Tools: LaTeX, Jupyter/Colab, Git & GitHub, Linux (Ubuntu), model serving and deployment.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Huawei Certified ICT Professional — Artificial Intelligence.
- Huawei Certified ICT Associate — Artificial Intelligence; and — Cloud Services.
- TensorFlow Developer Certificate (Coursera).
- Deep Learning Specialization — Andrew Ng, DeepLearning.AI (Coursera).
- ASOM AI Machine Learning Summer School (online), Saudi Arabia (2023); Master Class on Artificial Intelligence, Pantech Solutions, India (2022).

PROFESSIONAL MEMBERSHIPS

- Nigerian Computer Society (NCS) — Member.
- Data Science Nigeria — Member.
- Nigerian Society of Engineers (NSE) — Graduate Member.

REFERENCES: Available upon request.